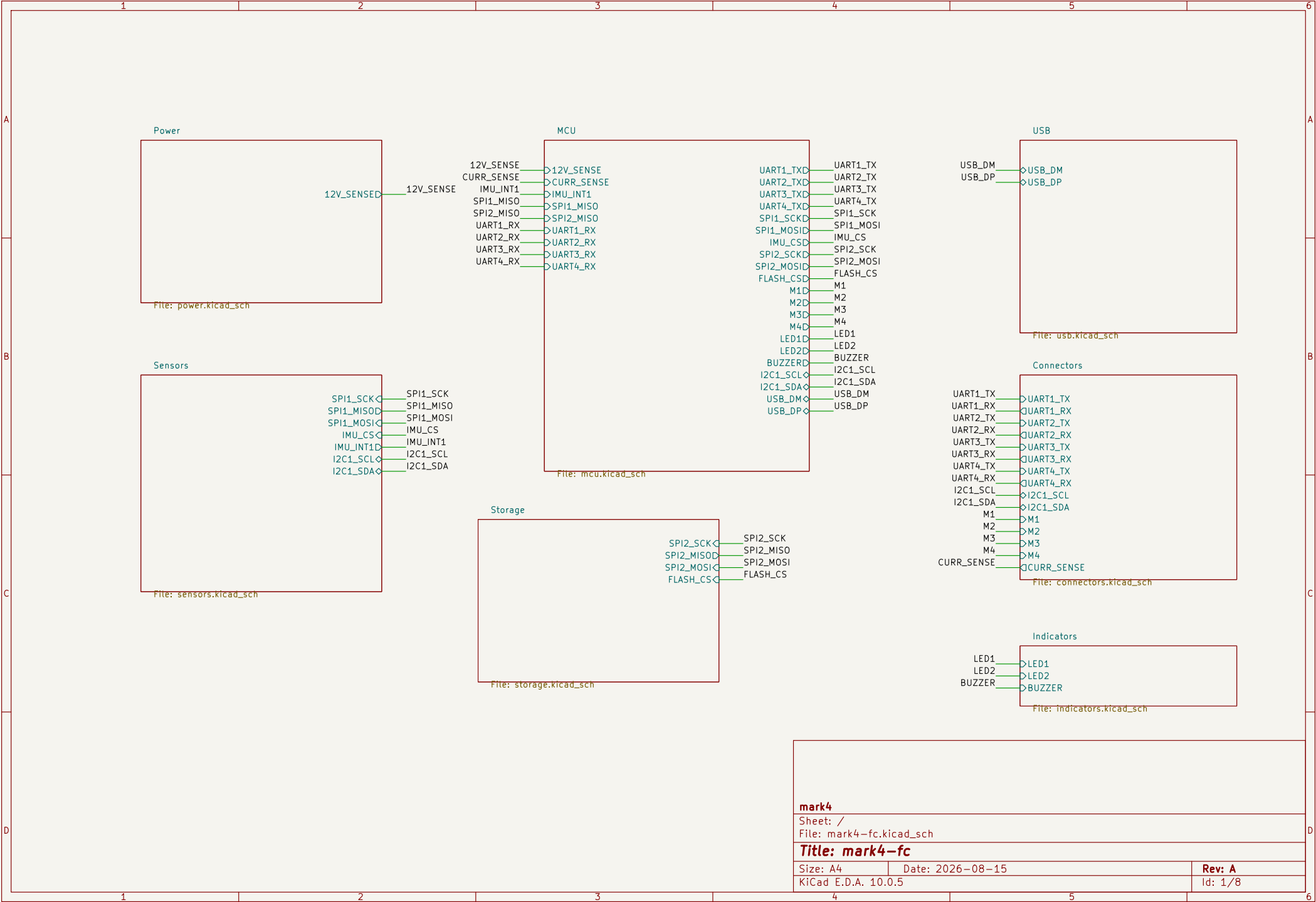
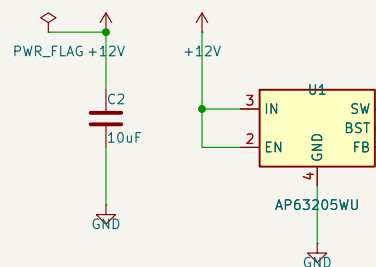


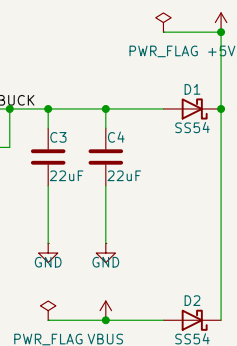


12 V input and 5 V buck (PWR-1)

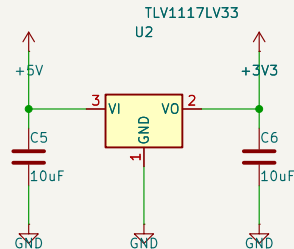


U1 is the fixed 5 V version: FB is tied to the buck output (5V_BUCK) per the datasheet, and EN is tied to VIN (always on). OR stage: D1 also blocks the AP63205 high-side body diode, so USB power on the shared rail cannot back-feed the 12 V net (PWR-3).

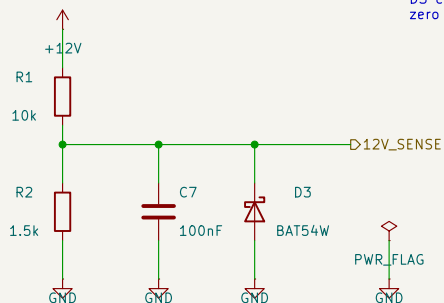
5 V OR-ing (PWR-3)



3V3 LDO (PWR-2)



12 V rail sense to ADC (PWR-4)



$R1/R2 = 10k/1.5k$, ratio $1.5/11.5 = 0.1304$. Tap = 1.57 V at 12 V nominal, and reaches 3.3 V (VDDA) only at a 25.3 V rail, so the ADC input stays below VDDA for any plausible transient on the ESC 12 V rail. D3 clamps negative excursions: DS11853 Table 60 allows zero negative current injection on PC0..PC3.

mark4

Sheet: /Power/
File: power.kicad_sch

Title: mark4-fc - Power

Size: A4 Date: 2026-08-15

KiCad E.D.A. 10.0.5

Rev: A

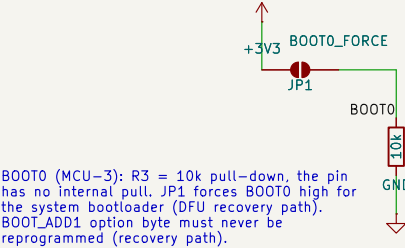
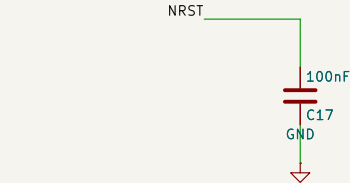
Id: 2/8

MCU – STM32F722RET6 (MCU–1). Pin assignment per docs/aio–board–pinmap.md

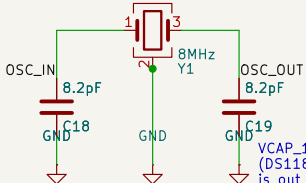
Decoupling per the pin map summary: C8–C11 = one 100 nF per VDD pin (19, 32, 48, 64), each placed at its pin on the PCB. C12 = 4.7 uF bulk on 3V3 near the MCU. C13 = 100 nF on VBAT (pin 1, tied to +3V3; no coin cell, no RTC backup).

VDDA (pin 13) is tied to +3V3 directly – NO series resistor:
VDDA is the ADC reference, an IR drop is a reference offset.
C14 100 nF + C15 1 uF at the pin (DS11853 Figure 28).
VSSA (pin 12) joins the ground plane at one point near the MCU.

NRST (pin 7): 100 nF at the pin (DS11853 Figure 47).
Internal pull–up 30–50k, no external one.
Not 5 V tolerant (4.0 V max) – exposed on J1 pin 5.



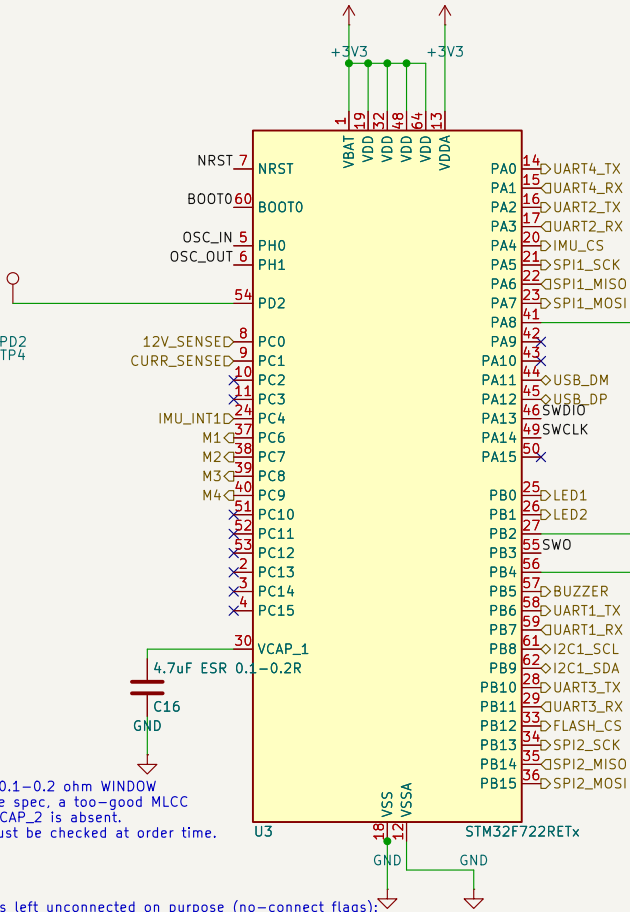
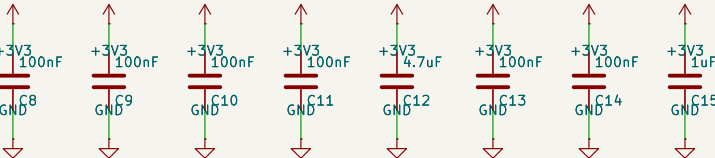
BOOT0 (MCU–3): R3 = 10k pull–down, the pin has no internal pull. JP1 forces BOOT0 high for the system bootloader (DFU recovery path).
BOOT_ADD1 option byte must never be reprogrammed (recovery path).



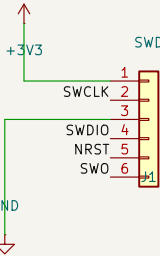
VCAP_1 (pin 30): 4.7 uF to VSS, ESR in the 0.1–0.2 ohm WINDOW (DS11853 Table 19 – the floor is part of the spec, a too–good MLCC is out of spec). Only VCAP pin on LQFP64. VCAP_2 is absent.
BOM TODO: no LCSC part chosen yet, ESR must be checked at order time.

HSE 8 MHz crystal (MCU–4): PLL /4 x216 /2 = 216 MHz sysclk, /9 = exact 48 MHz USB clock.
C18/C19 = 2 x (CL 9 pF – 5 pF stray) = 8.2 pF, NP0 only; trim 6.8/10 pF on the bench if the layout runs long.
Layout: guard ground ring around Y1 and its caps stitched to the MCU VSS, PH0/PH1 loop as short as possible, nothing routed under the crystal body.

Free pins left unconnected on purpose (no–connect flags):
PC13, PC14, PC15 (backup domain, no LSE fitted). PC2, PC3 (spare ADC inputs).
PA9 (OTG_FS_VBUS – sensing stays disabled, firmware forces B–valid). PA10 (OTG_FS_ID, device–only board). PA15, PC10, PC11 (UART escape route), PC12.
They stay reachable at the package if a respin ever needs them.



SWD header (MCU–2 / CON–4): 2.54 mm 1x6 through–hole, mark1 Prog pinout
1 = 3V3, 2 = SWCLK, 3 = GND, 4 = SWDIO, 5 = NRST, 6 = SWO (PB3);
NOT assembled: hand–populated with 3 pins (GND, SWCLK, SWDIO); all 6
positions are routed so the 1:1 cable option stays open. No BOM line.
SWD nets stay local to this sheet.



Test points (pin map section 4 recommendation):
TP1 = PA8 (MCO_1, clock bring–up probe), TP2 = PB2,
TP3 = PB4 (NJTRST at reset), TP4 = PD2 (only GPIO pin).

Not on this sheet: the I2C1 4.7k pull–ups on PB8/PB9 (sensor sheet), the 10k pull–up on
FLASH_CS/PB12 (storage sheet, the W25Q /CS must track VCC at power–up), the 12 V divider
and its Schottky feeding 12V_SENSE (power sheet, PWR–4), and the USB 90 ohm pair plus ESD
array on USB_DM/USB_DP (USB sheet, CON–5, no series resistors).
PA4 and PA5 are TTA pins (3.3 V max): nothing 5 V may ever touch IMU_CS or SPI1_SCK.

mark4

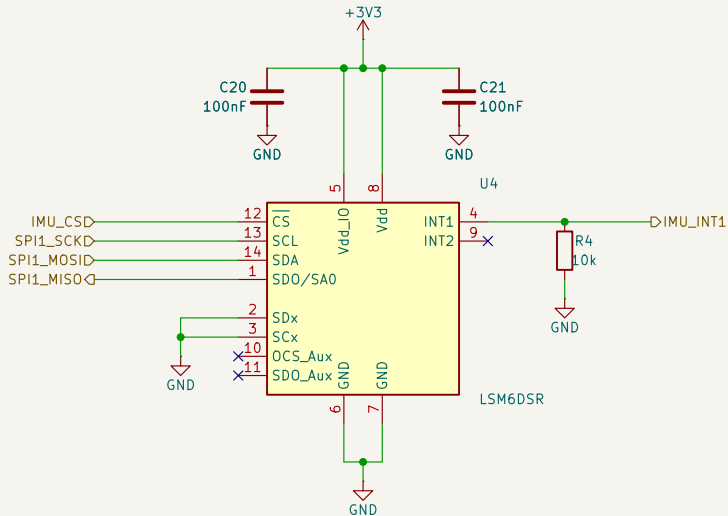
Sheet: /MCU/
File: mcu.kicad_sch

Title: mark4–fc – MCU

Size: A3 Date: 2026–08–15
KiCad E.D.A. 10.0.5

Rev: A
Id: 3/8

IMU – LSM6DSR on SPI1, dedicated CS (SNS–1)

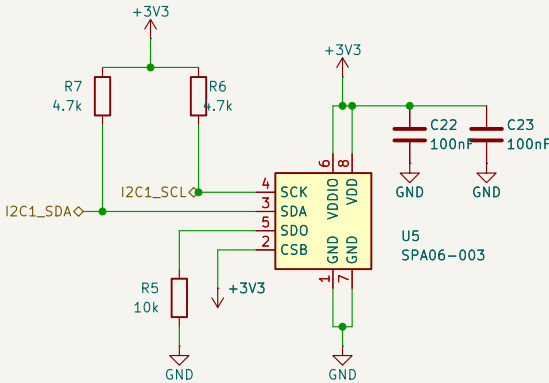


SNS–1 SPI Mode 1: SDx and SCx tied to GND select the 4–wire SPI primary interface. OCS_Aux and SDO_Aux stay UNCONNECTED (soldered, no net) – they carry internal pull–ups, so grounding them burns current and is out of spec. INT2 unused, no–connect flag. SPI1 clock 10 MHz max; F722 SPI1 prescaler /16 gives 6.75 MHz. FIRMWARE, first writes after reset: I2C_disable = 1 (CTRL4_C) and I3C_disable = 1. With CS high the I2C/I3C slave is otherwise live on the SPI lines.

SNS–2 INT1 is an interface strap sampled at IMU power–up (high = I3C–only mode). NO pull–up on this net, EVER. R4 10k pull–down is the optional safe default; the MCU pin stays input / low until the IMU is up. Routing INT1 to an EXTI–capable pin is mandatory, using it is optional (the loop may stay timer–paced).

SNS–5 IMU placement: board center, stiffest area, axes aligned with the board edges and marked on silkscreen, continuous ground plane underneath. C20 / C21 100 nF at the Vdd_IO / Vdd pins. The ESC silicone grommets are the vibration isolation (MEC–1).

Barometer – SPA06–003 on I2C1, addr 0x76 (SNS–3)



SNS–4 baro placement: pressure port facing UP, no silkscreen and no conformal coating on the port, room for an open–cell foam patch over it, away from the regulators (thermal gradient = drift) and away from the prop–wash board edges.

SNS–3 rate ceiling from the datasheet (Rate × Time < 1 s); 25 Hz allows 16x pressure oversampling, 50 Hz caps it at 8x. CSB tied to +3V3 (= VDDIO level) selects I2C. SDO to GND through R5 10k gives address 0x76; a resistor rather than a short keeps SDO–as–interrupt available.

R6 / R7 4.7k are the ONLY I2C1 pull–ups on the board. They live on this sheet because the barometer is the primary I2C user; the GPS connector (CON–2) shares the same bus and must NOT carry a second pair.

mark4

Sheet: /Sensors/
File: sensors.kicad_sch

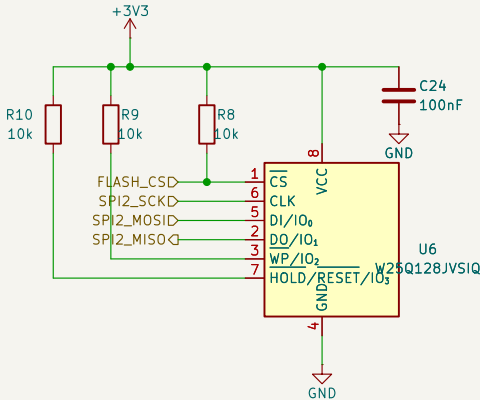
Title: mark4–fc – Sensors

Size: A3 Date: 2026–08–15
KiCad E.D.A. 10.0.5

Rev: A

Id: 4/8

Blackbox storage – W25Q128JV on its own SPI2 bus (STO–1)



STO–1 strapping. R8 10k on /CS: the die has no internal pull–up and the datasheet requires /CS to track VCC at power–up, when the MCU pin is still high–Z. R9 / R10 10k on /WP (IO2) and /HOLD (IO3): NEVER hard ties – the IQ order code ships with QE=1, which repurposes both pins as IO2/IO3, and the datasheet explicitly forbids direct supply or ground ties in that state. Own SPI bus, never shared with the IMU: logging writes must not contend with sensor reads. Plain SPI only, QUADSPI dropped (STO–3).

FOOTPRINT: the KiCad symbol default (SOIC–8 3.9x4.9 mm) is WRONG for the JVS1Q 208–mil wide body. Overridden to mark4–fc:SOIC–8_5.28x5.28mm_P1.27mm.

STO–5 throughput – why the bus is never the constraint, and the FIRMWARE rule. Required sustained rate is 32.5 kB/s (65–byte record at 500 Hz, STO–2). The W25Q128JV programs a 256–byte page in 0.7 ms typ / 3 ms max = 366 kB/s typ and 85 kB/s worst case, both above the requirement. SPI clock is irrelevant to writes (50 MHz ceiling for plain 03h reads, 27 MHz chosen on SPI2). What matters is ERASE latency: a 4 KB sector erase stalls the die 45 ms typ, up to 400 ms max = 13 kB of backlog at the required rate. FIRMWARE MUST therefore buffer records in a RAM ring of 16 KB or more (cheap out of 256 KB) and erase AHEAD of the write head.

Sizing (STO–2): 16 MB is about 8 minutes of armed logging: logging only while armed is the intended policy. W25Q256JVS1Q (32 MB, same footprint) MAY be substituted at order time. Retrieval over USB CDC (STO–4), no SD card.

mark4

Sheet: /Storage/
File: storage.kicad_sch

Title: mark4–fc – Storage

Size: A3

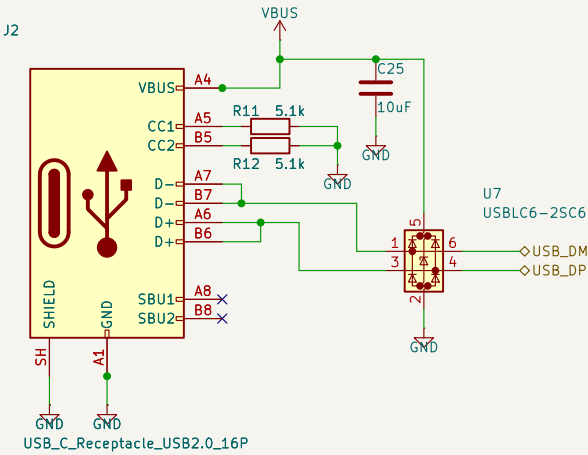
Date: 2026–08–15

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Id: 5/8

USB (CON-5) – USB-C 2.0 receptacle, UFP only

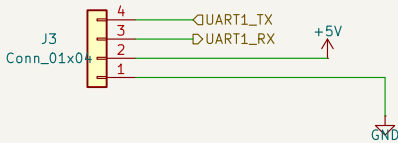


Notes:

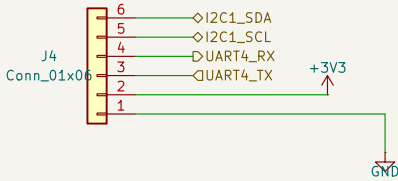
- Route USB_DP / USB_DM as a 90 ohm differential pair at layout (CON-5).
- No series resistors on D+/D- and no VBUS divider; firmware forces B-valid (pinmap PA9 row).
- R11 / R12: one 5.1k pulldown per CC pin, NEVER shared – this is a UFP-only port.
- C25 is local VBUS bulk near the receptacle; VBUS is the bench supply path (PWR-3).
- OR-ed with the buck output on the Power sheet.
- SBU1 / SBU2 unused (USB 2.0 only); shell and GND pins go straight to the GND plane.
- U7 sits between the receptacle and the MCU (flow-through): I/O1 = D-, I/O2 = D+.

External connectors – all JST–SH 1.00 mm, SIDE entry (SM..B–SRSS–TB)

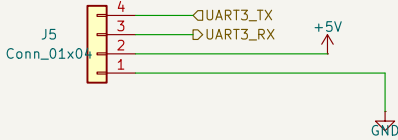
J3 – CON–1 telemetry UART (USART1): 1 GND, 2 +5V, 3 RX, 4 TX



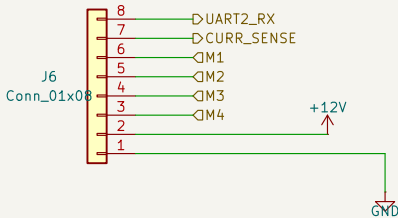
J4 – CON–2 GPS / external mag: 1 GND, 2 +3V3, 3 TX, 4 RX, 5 SCL, 6 SDA



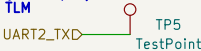
J5 – CON–3 spare UART (USART3, RC receiver)
1 GND, 2 +5V, 3 RX, 4 TX



J6 – ESC–3 4–in–1 ESC connector (pinout FROZEN by the spec)
1 GND, 2 +12V, 3 M4, 4 M3, 5 M2, 6 M1, 7 CURR, 8 TLM



- Notes:
- TX/RX are BOARD–side names: the pin named TX carries the board’s UART*_TX net. Check against the mark1 cables before crimping.
 - J3 (CON–1) and J5 (CON–3) are the same part; J5 may be DNP (SRC–3) until a real radio arrives.
 - J4 (CON–2) pin order was chosen at schematic time – the spec does NOT freeze it. Freeze this table before making the cable. The I2C1 pull–ups live on the Sensors sheet, so there are none here.
 - J6 (ESC–3) is the BOARD’S MAIN POWER INPUT on pin 2 (+12V, PWR–1). The pinout is frozen by the spec; verify the actual 4–in–1 harness wire order against this table before crimping the cable.
 - J6 pin 7 CURR is the analog current–sense fallback (PWR–4 / SRC–3); pin 8 TLM is the ESC telemetry UART RX (ESC–2). USART2_TX has no connector pin, only the TP5 pad.
 - Connectors sit on the board edges with the openings outward, side entry (MEC–2).
 - All hierarchical nets leave this sheet to the MCU sheet; nothing is wired between sheet blocks in the root.



mark4

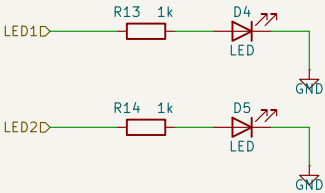
Sheet: /Connectors/
File: connectors.kicad_sch

Title: mark4–fc – Connectors

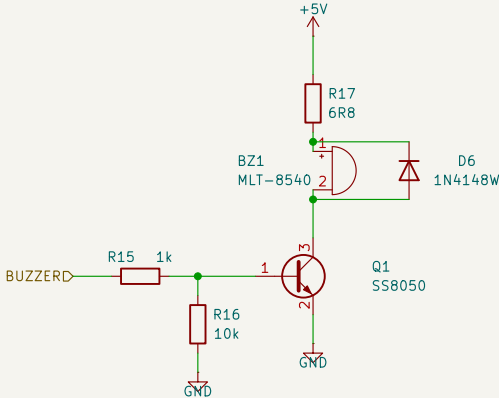
Size: A3	Date: 2026–08–15	Rev: A
KiCad E.D.A. 10.0.5		Id: 7/8

Indicators – status LEDs (IND–1) and buzzer driver (IND–2)

IND–1 – two status LEDs



IND–2 – buzzer, low–side driver



- Notes:
- IND–1: D4 / D5 are a BOM gap – the LED reference is not chosen yet (known).
R13 / R14 = 1k: at 3V3 that is about 1–2 mA depending on the LED Vf.
Adjust the value at order time once the LED is picked.
 - IND–2: passive magnetic buzzer, PB5 = TIM3_CH2 so tones are free in firmware.
 - R17 = 6R8 (decided 2026–08–15): the MLT–8540 coil is a 3.6 V class part on a post–OR rail near 4.6 V. 6.8 ohm drops about 1.5 V at the ~220 mA coil current and also protects the coil from a firmware bug leaving PB5 high, which a duty cap would not. 0805 for the dissipation; the LCSC part number is a TODO.
 - R15 = 1k base resistor gives deep saturation at ~220 mA.
 - R16 = 10k base–to–GND keeps the buzzer silent while PB5 is high–Z at reset.
 - D6 (1N4148W) is across the coil, cathode to the 5 V side.

mark4

Sheet: /Indicators/
File: indicators.kicad_sch

Title: mark4–fc – Indicators

Size: A3
Date: 2026–08–15

Rev: A
Id: 8/8